Measuring Term Premia for the Euro Area

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1 Introduction

Nominal yields at different maturities can be decomposed in two terms, capturing respectively the expectations of future monetary policy and the term premium. The first term can be represented as the market expectations on the average short rate (the monetary policy rate) over the maturity of the bond, while the term premium is a risk premium component representing the additional compensation required by the investor to be indifferent between a buy-and-hold strategy in the longer maturity bond and a roll-over investment strategy in the sequence of monetary policy rates. As Europe features a common monetary policy, term premia on bonds issued by different countries are an immediate gauge of the heterogeneity in the transmission mechanism of monetary policy in Europe and of the need for intervention to protect the Euro area against it. Given the observation of the yields on bonds at different maturities, the derivation of term premia requires a proxy for the average future short rate over the maturity of the bond. This website provides first historical estimates for the 10-year term premia from 2000 onwards for Germany, France, Spain and Italy using the consensus forecasts and a simple autoregressive model for the change in monetary policy rates to predict monetary policy rate in real time in each period. It then allows to compute current value of the term premia using the two historical benchmarks and forecasts for the future monetary policy rates inputed by the user. Many different models have been proposed in the literature to predict monetary policy and decompose yields to maturity into a term premium and an expected monetary policy components (see, for example, Favero et al. (2022)), and the term premium can be further decomposed in compensation for real rate risk and for inflation (see, for example, Haubrich et al. (2012),Berardi (2022)). Most macro-finance models of the term structure mostly belong to the class of Affine Term Structure Models (see, for example,(Ang and Piazzesi, 2003), (Diebold et al.,

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2005)). These models were originally thought for stationary processes in yields, as the yield dynamics is modelled via a VAR in some factors, partially or totally extracted form the Term Structure, and VAR models are designed for forecasting stationary processes. The factor dynamics also drives the price of risk and holding period returns. The potential problem with this approach is that the presence of stochastic trend in yields has several negative consequences for this approach. VARs are inappropriate for long-run forecasting of non-stationary data, biased forecast of the dynamics of short-term rates will affect the measurement of term premia and the non-stationatity of factors might result in non-stationarity of term premia, which is counterfactual as holding period returns are stationary. Exploiting the availability of a full zero-coupon equivalent term structure for Germany, the website also allows to evaluate term premia derived from consensus forecasts of the monetary policy rates with term premia derived from a standard ATSM model (Adrian et al. (2013)).

2 A General Framework

Consider zero-coupon bonds and define the relationship between price and yield to maturity of a zero-coupon bond as follows:

$$P_{t,T} = \frac{1}{(1 + Y_{t,T})^{T-t}}, \tag{1}$$

where $P_{t,T}$ is the price at time t of a bond maturing at time T, and $Y_{t,T}$ is yield to maturity. Taking logs we have:

$$p_{t,T} = -(T - t)\, y_{t,T}, \tag{2}$$

The one-period return of holding form period t to period t+1 a bond with maturity t+T is therefore:

$$r^T_{t,t+1} = p_{t+1,T} - p_{t,T}, \tag{3}$$

By applying the no-arbitrage condition to an investment of one-period in the one-period bond and an investment of one-period in the T-period bond we have :

$$E_t(r^T_{t,t+1} - r^1_{t,t+1}) = E_t(r^T_{t,t+1} - y_{t,t+1}) = \phi^T_{t,t+1}$$

where $\phi^T_{t,t+1}$ is the one-period risk premium on a one-period investment in the T-period bond.

Solving forward the difference equation $p_{t,T} = p_{t+1,T} - r^T_{t,t+1}$, we have :

$$
\begin{aligned}
y_{t,T} &= \frac{1}{(T-t)} \sum_{i=0}^{T-1} E_t \left(r^T_{t+i,t+i+1} \right) \\
&= \frac{1}{(T-t)} \sum_{i=0}^{T-1} E_t \left(y_{t+i,t+i+1} + \phi^T_{t+i,t+i+1} \right)
\end{aligned}
$$

Which provides the decomposition of nominal yields at different maturities into the market expectations on the average short rate (the monetary policy rate) over the maturity of the bond, and a risk premium component, the term premium, representing the additional compensation required by the investor to be indifferent between a buy-and-hold strategy in the longer maturity bond and a roll-over investment strategy in the sequence of monetary policy rates.The term premium over the residual life of the bond is the average of the sequence of the one-period future term premia.

By subtracting from the left-hand side and the right-hand side of the equation above the one-period rate, we have the following equation for the term spread:

$$
y_{t,T} - y_{t,t+1} = \sum_{i=1}^{T-t-1} \left(1 - \frac{i}{T-t} \right) E_t \Delta y_{t+i,t+i+1} + \frac{1}{T-t} \sum_{i=1}^{T-t-1} \phi^T_{t+i,t+i+1} \tag{4}
$$

3 Term Premia using Consensus Forecasts and a Naive Predictive Model

Equation (4) can be used to decomposed yields in an expected monetary policy component and a term premium given an estimate of the future expected future path for short-term rates $E_t \Delta y_{t+i,t+i+1}$ We implement the decomposition on the data by using the 3-month common short-rate for the Euro Area, and the yields on the 10-year government bond yields for the term spread. To estimate the future path of short-term rates we use two approaches -

1. Consensus forecasts -
   The ECB's Professional Forecasters Survey gives the average forecasts for the expected future short-term rates. These can be directly used to estimate the expected path of short-term rates..

2. Naive Predictive Model -

$$\Delta y_{t+1,t+2} = \rho \Delta y_{t,t+1} + \epsilon_{t+1}$$

where the autoregressive parameter ρ is estimated using the historical data available.

4 Term Premia on German 10Y bonds using Affine Term Structure model

Affine models of the term structure of interest rates are a popular way of determining the term premia. The expectation of the future path of short rates can be extracted from these term structure models. The affine models typically use state variables (latent factors) to model the shocks that drive the economy. The key assumptions are:

1. The pricing kernel is exponentially affine in the state variables
2. Market prices of risk are affine in the state variables
3. The innovations to state variables and log yield observation errors are conditionally Gaussian.

These assumptions lead to two key equations - the measurement equation where the yields are affine in the state variables and the transition equation which describes the evolution of the state variables. This model can be solved using maximum likelihood estimation.

Adrian et al. (2013) proposed an interesting regression based solution for affine term structure models. We implement their model (ACM) to compute the term premia for Germany.

4.1 Expected Returns Dynamics in the ACM-ATS Model

The dynamics of a K $\times$ 1 vector of state variables Xt evolve according to the following vector autoregression (VAR):

$$X_{t+1} = \mu + \Phi X_t + v_{t+1} \tag{5}$$

$$v_{t+1} | \{X_s\}_{s=0}^{t} \sim \mathcal{N}(0, \Sigma), \tag{6}$$

The market price of risk are assumed to be in the following affine form:

$$\lambda_t = \Sigma^{-1/2}(\lambda_0 + \lambda_1 X_t), \tag{7}$$

The assumption of no-arbitrage implies that there exists a pricing kernel M_t such that:

$$P_t^{(n)} = E_t\left(M_{t+1} P_{t+1}^{(n)}\right), \tag{8}$$

Assuming the pricing kernel, M_{t+1}, is exponentially affine:

$$M_{t+1} = \exp\left(-r_t - \frac{1}{2}\lambda_t'\lambda_t - \lambda_t'\Sigma^{-1/2} v_{t+1}\right), \tag{9}$$

where $r_t = -\ln P_t^{(1)}$ is the continuously compounded risk-free rate. The excess log returns are given by:

$$rx_{t+1}^{(n-1)} = \ln P_{t+1}^{(n-1)} - \ln P_t^{(n)} - r_t, \tag{10}$$

By substituting in the no-arbitrage condition the expression for the pricing kernel M_{t+1}, and by using the definition of excess returns we have

$$1 = E_t\left[exp\left(rx_{t+1}^{(n-1)} - \frac{1}{2}\lambda_t'\lambda_t - \lambda_t'\Sigma^{-1/2} v_{t+1}\right)\right], \tag{11}$$

taking logs we have

$$0 = log\left[E_t\left[exp\left(rx_{t+1}^{(n-1)} - \frac{1}{2}\lambda_t'\lambda_t - \lambda_t'\Sigma^{-1/2} v_{t+1}\right)\right]\right], \tag{12}$$

Assume $\left\{rx_{t+1}^{(n-1)}, v_{t+1}'\right\}$ are jointly normally distributed, then we have:

$$\begin{aligned} log \quad & \left[E_t\left[exp\left(rx_{t+1}^{(n-1)} - \frac{1}{2}\lambda_t'\lambda_t - \lambda_t'\Sigma^{-1/2} v_{t+1}\right)\right]\right] \\ = \quad & E_t\left(rx_{t+1}^{(n-1)}\right) + \frac{1}{2} Var_t\left(rx_{t+1}^{(n-1)}\right) - \frac{1}{2}\lambda_t'\lambda_t + \frac{1}{2}\lambda_t'\lambda_t \\ - \quad & Cov_t\left[rx_{t+1}^{(n-1)}, v_{t+1}'\Sigma^{-1/2}\lambda_t\right] \end{aligned} \tag{13}$$

We can then write:

$$E_t\left(rx_{t+1}^{(n-1)}\right) = Cov_t\left[rx_{t+1}^{(n-1)}, v_{t+1}'\Sigma^{-1/2}\lambda_t\right] - \frac{1}{2} Var_t\left(rx_{t+1}^{(n-1)}\right), \tag{14}$$

and denote $\beta_t^{(n-1)'}$ as:

$$\beta_t^{(n-1)'} = \text{Cov}_t \left[rx_{t+1}^{(n-1)}, v'_{t+1} \right] \Sigma^{-1}, \tag{15}$$

By substituting from (15) into (14), using (7), we have:

$$E_t \left(rx_{t+1}^{(n-1)} \right) = \beta_t^{(n-1)'} (\lambda_0 + \lambda_1 X_t) - \frac{1}{2} Var_t \left(rx_{t+1}^{(n-1)} \right), \tag{16}$$

The unexpected excess return can be decomposed in a component that is correlated with v_{t+1} and another component which is orthogonal:

$$rx_{t+1}^{(n-1)} - E_t \left(rx_{t+1}^{(n-1)} \right) = \beta_t^{(n-1)'} v_{t+1} + e_{t+1}^{(n-1)}, \tag{17}$$

Under the assumption that the return pricing error are i.i.d. with variance σ^2 and that β_t is constant, the generating process for log excess returns becomes:

$$rx_{t+1}^{(n-1)} = \underbrace{\beta^{(n-1)'}(\lambda_0 + \lambda_1 X_t)}_{\text{Expected Return}} - \underbrace{\frac{1}{2}(\beta^{(n-1)'} \Sigma \beta^{(n-1)} + \sigma^2)}_{\text{Convexity Correction}} + \underbrace{\beta^{(n-1)'} v_{t+1}}_{\text{Return Innovation}} + e_{t+1}^{(n-1)}, \tag{18}$$

Stacking (18) across N maturities and T time-periods we have the following matrix-form representation:

$$rx = \beta'(\lambda_0 \iota_T' + \lambda_1 X_-) - \frac{1}{2}(B^* \text{vec}(\Sigma) + \sigma^2 \iota_N)\iota_T' + \beta' V + E, \tag{19}$$

rx is a $N \times T$ matrix of excess returns.
$\beta = [\beta^{(1)}, \beta^{(2)}, \ldots, \beta^{(N)}]$ is a $K \times N$ matrix of factor loadings.
ι_T and ι_N are $T \times 1$ and $N \times 1$ vectors of ones.
$X_- = [X_0, X_1, \ldots, X_{T-1}]$ is a $K \times T$ matrix of lagged pricing factors.
$B^* = [\text{vec}(\beta^{(1)}\beta^{(1)'}) \ldots \text{vec}(\beta^{(N)}\beta^{(N)'})]$ is an $N \times K^2$ matrix.
V is a $K \times T$ matrix, and E is an $N \times T$ matrix

4.2 Parameters' Estimation

The estimation is done in 3 steps

1. Construct the pricing factors (X) from principle component analysis (PCA) of the yields. Estimate Eq (6) using OLS, decomposing the pricing factors into predictable components and factor innovations $\hat{V}$.

2. Regress excess returns on a constant, lagged pricing factors and contemporaneous pricing factor innovations according to

$$rX = a\iota_T' + \beta'\hat{V} + cX_- + E, \tag{20}$$

We can compute B^* from β and σ^2 from the residuals E.

3. From Eq (19), we can see $a = \beta'\lambda_0 - \frac{1}{2}(B^*\text{vec}(\Sigma) + \sigma^2\iota_N)$ and $c = \beta'\lambda_1$. From these we get the estimators -

$$\hat{\lambda}_0 = (\hat{\beta}\hat{\beta}')^{-1}(\hat{a} + \frac{1}{2}(\hat{B}^*\text{vec}(\hat{\Sigma}) + \hat{\sigma}^2\iota_N)) \tag{21}$$

$$\hat{\lambda}_1 = (\hat{\beta}\hat{\beta}')^{-1}\hat{\beta}\hat{c} \tag{22}$$

4.3 Affine Yields, Risk Neutral Yields and Term Premia

The model assumptions imply -

$$\ln P_t^n = A_n + B_n' X_t + u_t^n, \tag{23}$$

Further, the risk-free rate r_t can be expressed as a linear function of the underlying factors -

$$r_t = \delta_0 + \delta_1' X_t + \epsilon_t, \tag{24}$$

We use OLS to get the estimators $\hat{\delta}_0$ and $\hat{\delta}_1$. Using (10) and (18)

$$\ln P_{t+1}^{(n-1)} - \ln P_t^{(n)} - r_t = \beta^{(n-1)'}(\lambda_0 + \lambda_1 X_t) - \frac{1}{2}(\beta^{(n-1)'}\Sigma\beta^{(n-1)} + \sigma^2) + \beta^{(n-1)'}v_{t+1} + e_{t+1}^{(n-1)}, \tag{25}$$

and we can obtain the price of a bond at any maturities by solving forward (25), using the terminal condition that the log of the price at maturity of any zero coupon bonds is zero and projecting future states on the current ones using the VAR, $X_{t+1} = \mu + \Phi X_t + v_{t+1}$. So for the one-period bond, by considering (23) and (24) we have:

$$A_1 = -\delta_0, B_1 = -\delta_1 \tag{26}$$

For the two-period bond:

$$\begin{aligned} \ln P_{t+1}^{(1)} - \ln P_t^{(2)} - r_t &= \beta^{(1)'}(\lambda_0 + \lambda_1 X_t) - \frac{1}{2}(\beta^{(1)'}\Sigma\beta^{(1)} + \sigma^2) + \beta^{(1)'}v_{t+1} + e_{t+1}^{(1)}, \\ \ln P_{t+1}^{(1)} &= -(r_{t+1}) = -\delta_0 - \delta_1'(\mu + \Phi X_t) - \delta_1' v_{t+1} \end{aligned}$$

By using the second equation in the first one, together with the VAR we have

$$\ln P_t^{(2)} = -\delta_0 - \delta_1' X_t - \delta_0 - \delta_1'(\mu + \Phi X_t) - \beta^{(1)'}(\lambda_0 + \lambda_1 X_t) + \frac{1}{2}(\beta^{(1)'}\Sigma\beta^{(1)} - \sigma^2) + \beta^{(1)'} v_{t+1} + e_{t+1}^{(1)}$$

In general, the model implies that bond prices depend linearly on the state vector:

$$p_t^n = A_n + B_n' X_t + u_t^n, \tag{27}$$

The model also implies cross-equation restrictions on the parameters A_n, B_n . Using equation (27) to derive excess returns, we have:

$$\begin{aligned} rx_{t+1}^{(n-1)} &= p_{t+1}^{(n-1)} - p_t^{(n)} - p_t^{(1)} \\ &= A_{n-1} + B_{n-1}' X_{t+1} + u_{t+1}^{n-1} - A_n - B_n' X_t + u_t^n + A_1 + B_1' X_t + u_t^1 \\ &= A_{n-1} + B_{n-1}'(\mu + \Phi X_t + v_{t+1}) + u_{t+1}^{n-1} - A_n - B_n' X_t + u_t^n + A_1 + B_1' X_t + u_t^1 \\ &= (B_{n-1}')v_{t+1} + (B_{(n-1)}'\Phi - B_n' + B_1')X_t + (A_{n-1} - B_{n-1}'\mu - A_n + A_1 + u_{t+1}^{(n-1)} - u_t^{(n)} + u_t^{(1)}) \end{aligned}$$

The return generating expression is also given by (18), which can be rewritten as:

$$rx_{t+1}^{(n-1)} = (\beta^{(n-1)'})v_{t+1} + (\beta^{(n-1)'}\lambda_1)X_t + (\beta^{(n-1)'}\lambda_0 - \frac{1}{2}\left(\beta^{(n-1)'}\Sigma\beta^{(n-1)} + \sigma^2\right) + e_{t+1}^{(n-1)})$$

For the one-period bond we have:

$$A_1 = -\delta_0, \quad B_1 = -\delta_1, \tag{28}$$

Matching the two equations for excess returns term-by-term we get the following:

$$B_{n-1}' = \beta^{(n-1)'}, \tag{29}$$

$$B_n' = B_{n-1}'(\Phi - \lambda_1) - \delta_1', \tag{30}$$

$$A_n = A_{n-1} + B_{n-1}'(\mu - \lambda_0) + \frac{1}{2}(B_{n-1}'\Sigma B_{n-1} + \sigma^2) - \delta_0, \tag{31}$$

Solving this system, we can use A_n and B_n in Eq (23) to get the model implied yields. Setting the market price of risk, i.e. λ_0 and λ_1, to zero and then solving the recursive equations, we can extract the risk neutral yields. We can compute the 10 year term premia by taking the difference between the model implied 10 year yield and the risk neutral yield.

4.4 Constructing Expected Path of Short Rates

Using (24) and (6) we can get the expected path of short (risk free rates) at time t for the next T periods.

$$E[r_{t+1}] = \delta_0 + \delta_1\mu + \delta_1\phi X_t, \tag{32}$$

$$\begin{aligned} E[r_{t+2}] &= \delta_0 + \delta_1\mu + \delta_1\phi E[X_{t+1}] && (33)\\ &= \delta_0 + \delta_1\mu + \delta_1\phi\mu + \delta_1\phi^2 X_t, && (34) \end{aligned}$$

$$\dots$$

$$E[r_{t+i}] = \delta_0 + \delta_1\mu + \delta_1\phi\mu + \delta_1\phi^2\mu + \ldots + \delta_1\phi^i X_t, \tag{35}$$

4.5 Data

Similar to Adrian et al. (2013), the zero-coupon yields are fitted using the Nelson-Siegel-Svensson (NSS) model. The NSS parameters, sourced from Deutsche Bundesbank, are used to calculate zero-coupon-rates for maturities from 1-120 months. From these yields we extract 5 principal components. The excess returns are calculated for 6, 18, 24, 36, 48, 60, 84 and 120 month maturities using the 1 month rate as the risk-free-return.

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